



Society of Petroleum Engineers

SPE-227501-MS

Coiled Tubing Operation and the Use of Non-Azimuthal Inversion of Ultra-Slim Multiple Propagation Resistivity

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227501-MS](https://doi.org/10.2118/227501-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

This paper explores the challenges and advancements in under-balanced coiled tubing drilling (UBCTD) operations for gas reservoirs, focusing on the use of non-azimuthal inversion of ultra-slim multiple propagation resistivity tools. The primary goal is to enhance geosteering capabilities and optimize well placement within gas sweet spot layers. Emphasis is placed on determining the distance of detection (DOD) and formation dip to achieve successful drilling of coiled tubing gas wells.

We review existing geosteering and monitoring technologies, with a focus on non-azimuthal inversion techniques for ultra-slim multiple propagation resistivity tools. The inversion process allows for better detection of formation dip and precise determination of DOD in challenging UBCTD environments. The technique is applied in combination with other technologies such as gamma ray measurements, cutting analysis, and real-time surface data to form a comprehensive approach. These methods are integrated to provide accurate and reliable data that support optimal lateral placement and efficient gas reservoir navigation.

The non-azimuthal inversion technique demonstrated significant advantages in determining DOD and accurately identifying formation dip. These results enable improved steering decisions in real time, ensuring that the coiled tubing is placed within the gas sweet spot layers. The combination of resistivity inversion and other measurement tools proves crucial in overcoming the limitations posed by the small size of the coiled tubing and dynamic hole conditions, ultimately leading to more successful wellbore placement and enhanced gas recovery.

This paper introduces a novel approach in the application of non-azimuthal inversion of ultra-slim multiple propagation resistivity to accurately determine DOD and formation dip. By integrating these inversion results with other geosteering technologies, this method improves the efficiency and success rate of drilling coiled tubing gas wells. The integration of these advanced technologies ensures optimal well placement, contributing to better reservoir navigation, increased gas recovery, and improved field development strategies.

Introduction

Underbalanced coiled tubing drilling (UBCTD) is an advanced drilling technique used in low pressure gas reservoirs development. UBCTD enhances reservoir connectivity while minimizing formation damage and lost circulation. Normally, in managed pressure drilling or overbalance drilling formation damages are common and are associated with drilling mud invasion into the formation. However, one of the challenges in UBCTD is accurate well placement and real-time reservoir evaluation due to the limitations of conventional Logging While Drilling (LWD) tools in ultra-slim hole coiled tubing environments.

Underbalanced drilling operation in low pressure gas reservoirs has been widely used. For example, an onshore Sajaa Field located in UAE targeting the carbonate Thamama Reservoir (Pruitt et al, 2004 and S. Abu Alsaud et al 2024). The primary objective of using underbalanced drilling includes to increase the gas reserves from the field that had been put on production since 1980, where the reservoir pressure was estimated to be in range of 1100 to 1900 psi. Fig.1 shows productivity gains of Sajaa Field where seven wells have been sidetracked with UBCTD, and all of these wells have shown significant gas production rate after intervention with UBCTD (Van der Werken et al., 2005 and S. Abu Alsaud et al 2024).

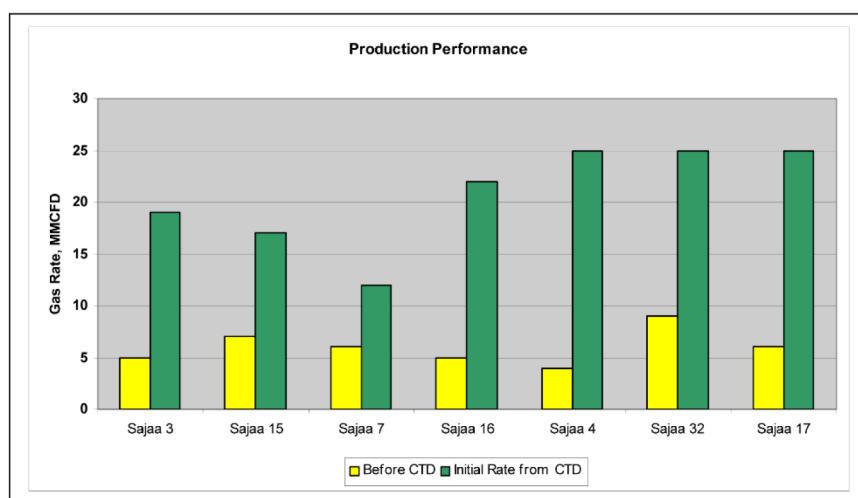


Figure 1—Productivity enhancement before and after sidetracking with UBCTD in the Sajaa onshore gas field located in Sharjah, United Arab Emirates, targeting the Thamama limestone (Pruitt et al, 2004 and S. Abu Alsaud et al 2024).

Ultra-slim multiple propagation resistivity technology with non-azimuthal inversion enables proactive reservoir navigation in UBCTD. Challenges to overcome is the limited LWD tools/ technologies for ultra-slim borehole. Traditional azimuthal resistivity tools provide directional resistivity measurements, but in UBCTD, where tool rotation is limited or absent, non-azimuthal inversion becomes essential. This approach reconstructs formation resistivity using multi-depth propagation data without requiring azimuthal segmentation.

The real-time non-azimuthal inversions, described as a multiparametric algorithm to describe complex models with multiple layers from resistivity measurements, which provides improved resistivity profiling and bed boundary detection. This paper discusses the integration of ultra-slim multiple propagation resistivity technology with non-azimuthal inversion to improve formation evaluation based on resistivity contrast to meet well placement objective in challenging drilling environments.

Geological Background

The studied formation is characterized by a complex depositional history and reservoir properties. The formation unconformably overlies older units, filling erosional topography created by tectonic activity. It

is primarily composed of fluvial to marginal marine sandstones, with evidence of glacial and periglacial influences in its lower sections.

The geological complexity of this formation is further compounded by significant structural variations and extensive erosional unconformities. Stratigraphically, the formation is composed of multiple key layers, each with unique depositional characteristics. A lower layer predominantly consists of glaciogenic sediments, which are overlain by a layer of widespread fluvial and deltaic deposits featuring interbedded sandstones, siltstones, and shales. The upper section reflects complex depositional patterns resulting from climatic shifts towards arid and semi-arid conditions (Al-Masrahy, 2017, 2020). This formation includes ephemeral lake deposits, tidal flats, eolian sands, and nearshore marine environments, leading to substantial heterogeneity in reservoir properties (Alshammery, 2017), (Figure 2).

Challenges and Objectives of the Well

The studied well encountered various geological and operational challenges due to the variable porosity and permeability of the formation, necessitating precise drilling and monitoring techniques to maximize reservoir contact and hydrocarbon extraction.

The reservoir in this area is very thick however there are resistivity variations within it which made it challenging to capture the detailed resistivity variations while resolving the formation apparent dip with limited LWD technologies for formation evaluation in ultra-slim borehole environment.

Key objectives for Well consist of following:

- Evaluation of resistivity data while drilling the slant well within the reservoir.
- Mapping of the boundaries with different resistivities.
- Estimation of formation apparent dip.
- Identification of the conductive layers within the reservoir.

These challenges and objectives emphasize the necessity of employing advanced workflows and data integration with the availability of limited technology to achieve successful drilling outcomes.

Technology Overview

UBCTD Drilling technology

UBCTD involves drilling with coiled tubing while maintaining underbalanced condition and is a breakthrough technology in developing mature and depleted reservoirs. The UBCTD applications restrict formation damage while drilling and let the effluent flow up to the surface to assess reservoir potential. An illustration of different drilling methods is shown in Figure 3 (Al-Omair, A., et al., 2011).

In the underbalance drilling technique, the density of the drilling fluid is kept lower than reservoir pressure to allow the reservoir fluids to enter the wellbore which allows drilling and production at the same time. Usually, nitrogen gas is used to lower the density of the mixture. Due to the inert behavior of nitrogen, it can be injected into any single-phase fluid (water, oil, diesel or any polymer-based mud) without risking the overall operation (Al Abdrabalnabi, H., et al., 2024; Bennion et al., 1996).

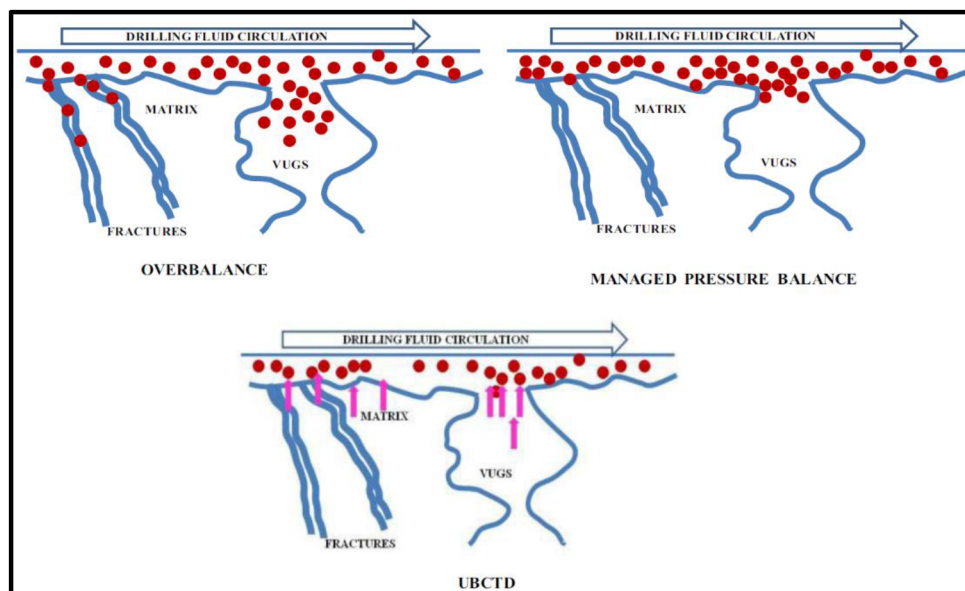


Figure 3—A visual explanation of different drilling methods OBD, MPD and UBCTD.

Ultra Slim Multi Propagation Resistivity Tool

Ultra-Slim Multi-Propagation Resistivity tool and its services have become integral part to modern Coiled Tubing Drilling (CTD) operations, particularly in mature fields where precise reservoir navigation is essential. The ultra-slim multiple propagation resistivity technology was developed in the late 1990s, implemented in the first generation of CTD BHAs and later transferred to the latest CTD BHA design. It is designed to fit within the limited diameter constraints of coiled tubing while delivering high-resolution resistivity measurements.

Ultra-Slim Multi-Propagation Resistivity tool operates by emitting electromagnetic waves at multiple frequencies, allowing for the differentiation of formation properties and fluid contents. It incorporates four quantitative resistivity measurements, based on two frequencies (2 MHz and 400 kHz) and two attenuations. The slim profile ensures minimal disruption to the drilling process while providing critical data for geosteering and formation evaluation (Krueger, S., et al., 2013).

The successful deployment of ultra-slim multiple propagation resistivity tool in CTD operations has been facilitated by advancements in Bottom Hole Assembly (BHA) designs. These BHAs integrate ultra-slim multiple propagation resistivity with other measurement tools, enabling comprehensive data acquisition without compromising the mechanical integrity of the drilling system. (Figure 4).

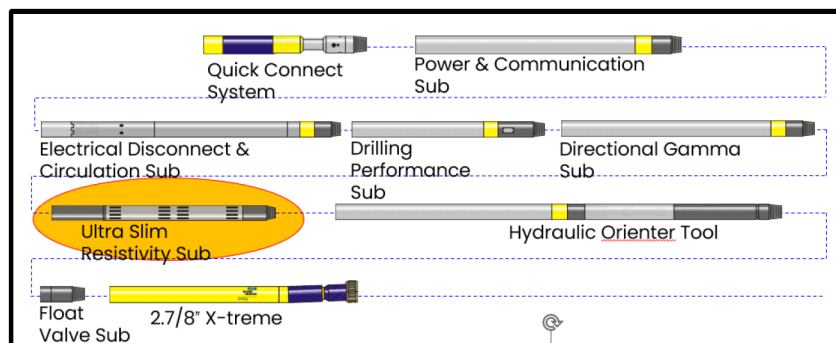


Figure 4—An illustration of CTD BHA with ultra-slim multiple propagation resistivity

Benefits of implementation of ultra-slim multiple propagation resistivity tools in UBCTD

- **Enhanced Reservoir Navigation:** Real-time resistivity measurements allow for precise wellbore placement within target zones, maximizing hydrocarbon recovery.
- **Improved Formation Evaluation:** Multi-propagation resistivity data provide insights into formation properties, aiding in the identification of productive intervals.
- **Operational Efficiency:** The slim design of the ultra-slim multiple propagation resistivity tools enables their deployment in smaller boreholes, reducing the need for larger drilling equipment and associated costs.

Technical Specifications of ultra-slim multiple propagation resistivity tool

- **Tool Sizes:** 3" Commonly.
- **Maximum Temperature:** Approximately 150°C (302°F)
- **Detection Depth:** Typically, up to 9 feet (2.7 meters)
- **Resistivity Range:** Capable of measuring from 0.1 to 3,000 ohm-m

Non-Azimuthal Inversion utilizing ultra-slim multiple propagation resistivity

Traditional azimuthal resistivity tools provide directional resistivity measurements. However, in coiled tubing drilling, where tool rotation is limited or absent, non-azimuthal inversion becomes essential. Ultra-slim multiple propagation resistivity tools with non-azimuthal inversion offer a novel method for deriving subsurface resistivity models using multi-frequency and multi-spacing data without requiring azimuthal segmentation.

Raw resistivity data is too complex to be interpreted. Multiple propagation resistivity curves are used as input for Non-Azimuthal Inversions. Using a 1D layered earth model, the inversion solves for true resistivity values by minimizing the difference between measured and modeled tool responses. This approach yields vertical resistivity profiles and boundary estimates and helps to prepare optimum earth model explaining all resistivity data.

Applications and Benefits of Non-Azimuthal Inversion

- **Accurate Boundary Detection:** Detects formation transitions based on resistivity contrast.
- **Improved Well Placement:** Enables proactive geosteering without azimuthal measurements.
- **Real-Time Formation Mapping:** Provides continuous formation resistivity profiles.
- **Capable of defining:** Formation dip and layers resistivity.

Other Monitoring Methods for UBCTD

Earlier, UBCTD operation relies greatly on biosteering to place the laterals in the desired layers. The biosteering application depends on the key microfossils located within the reservoir to detect the target layer and define drilling projection.

Productivity Index (PI) curve is another indicator for accurate placement of wellbore in the targeted layer and ensuring the production contribution of each layer. The PI curve is constructed against the drilling progress using the following equation:

$$PI = Q / \Delta P$$

Here, Q is the gas flow rate in million standard cubic feet per day (MMscfd) and ΔP is the differential pressure between reservoir pressure and the flowing bottom-hole pressure in psi. To ensure continuous contribution from the target layers, the PI curves are monitored closely. (Al-Omair, A., et al., 2011)

Pre-Well Planning and Feasibility Study

A thorough pre-well planning for Well was essential to understand the complex geological conditions expected within the target reservoir. The well plan was aimed to drill the 3.625" lateral section as slant well within the reservoir. The data derived from mother borehole Well_0 was utilized as main offset wells to create the RNS ultra-slim multiple propagation resistivity forward model considering other offset wells were 4-10 km away from the well.

Pre-well modeling showed that the reservoir is thick, with layers of varying resistivities within it with minor lithological changes. Higher resistivity zones are generally correlated with higher productivity in the studied formation. These different resistive layers within the reservoir posing a significant challenge to map the resistive boundaries precisely.

Based on the offset wells, observed general high resistivity environment (50-100 Ohmm) across the target reservoir. Some zones within the reservoir are relatively lower resistive (13-20 Ohmm). The low resistivity contrast as identified here through detailed multi-frequency resistivity modeling, reduces the non-azimuthal inversion depth of detection (DOD) to 5ft TVD. The pre-well scenario model with Non-Azimuthal Inversion showed that it can identify different zones based on their resistivity contrast within the tool DOI and map the zones thicknesses (Figure 6).

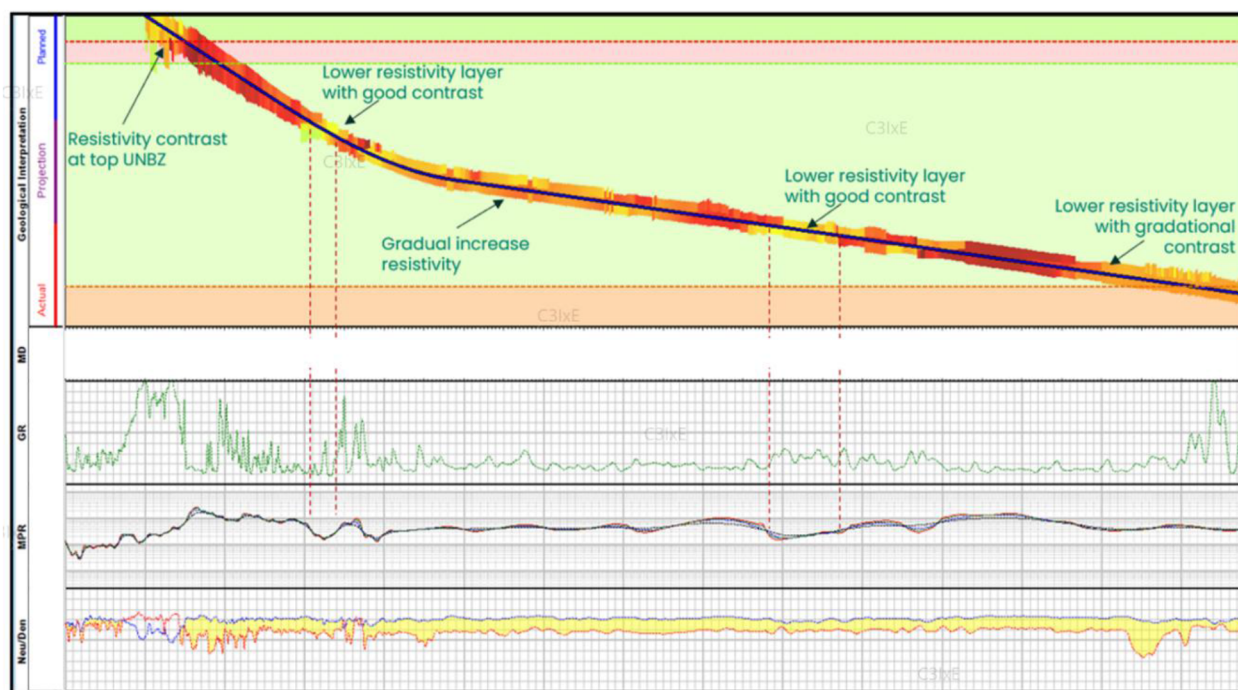


Figure 5—Pre-well modeling cross-section illustrating expected resistivity contrasts and anticipated geological structures derived from offset well data and inversion modeling.

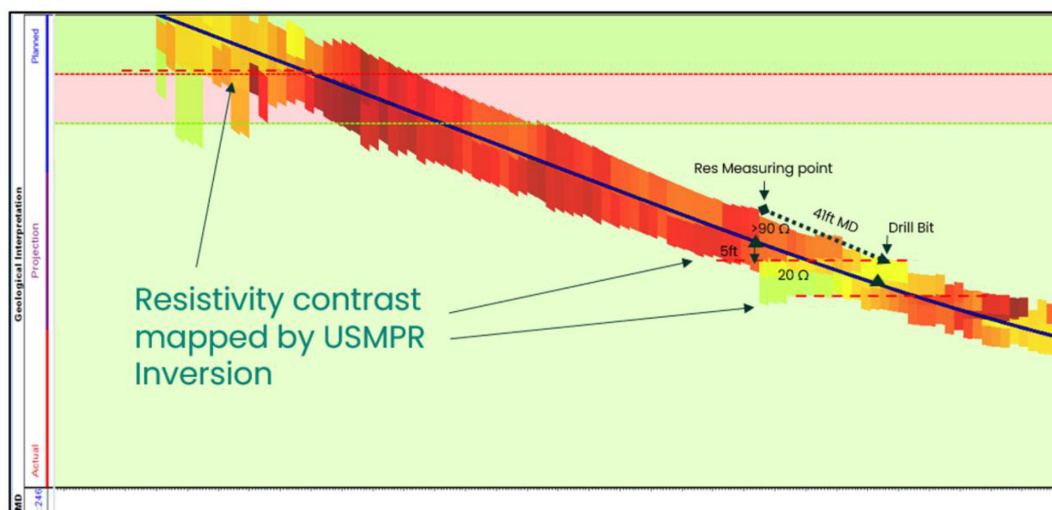


Figure 6—Pre-well modeling cross-section illustrating expected resistivity contrasts mapping through Non-Azimuthal Inversion.

Real-Time Monitoring and Operational Results

Real-time monitoring and operational adjustments were crucial to the successful execution of the Well. Drilling operations commenced with buildup section. 3" ultra-slim multiple propagation resistivity tool along with non-azimuthal inversion was implemented to evaluate resistivity data while drilling the slant well within the reservoir. Real-time monitoring and continuous data interpretation allowed the team to promptly respond to the complex geological conditions encountered.

Initial monitoring identified formation apparent dip angles varying from 1.2° up-dip to 0.8° down-dip along the well trajectory. Key operational adjustments included strategic inclination changes implemented at critical depths addressing the formation dip changes to meet the well objective to model the resistivity responses for entire reservoir.

The buildup section shown in the figure 7, clearly explaining the boundaries of different resistivities mapped by the non-azimuthal inversion while kept cutting down with high angle in the buildup section. A conductive layer was encountered in this section mapped by the non-azimuthal inversion as illustrated in Figure 7. Formation apparent dip was identified 1.2° up-dip in the section in general.

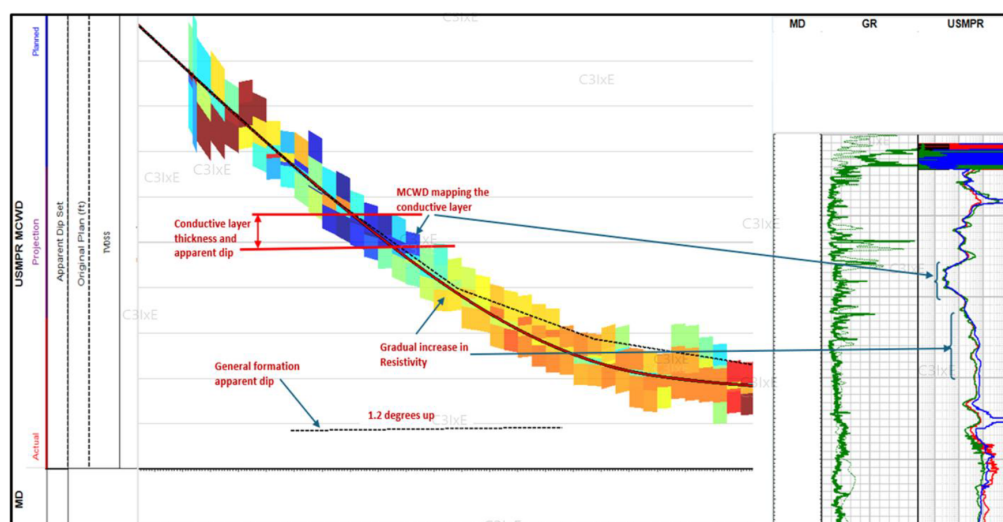


Figure 7—Real-time non-azimuthal inversion results for buildup section, illustrating mapping of different resistivity layers.

The slant section inversion results are shown in figure 8, illustrating the wellbore continued cutting down to different resistivity layer and these resistivity profiles were mapped successfully by the non-azimuthal inversion results within the DOD of 5ft TVD based on the resistivity contrast. Formation apparent dip was identified 0.8° down-dip in this section in general.

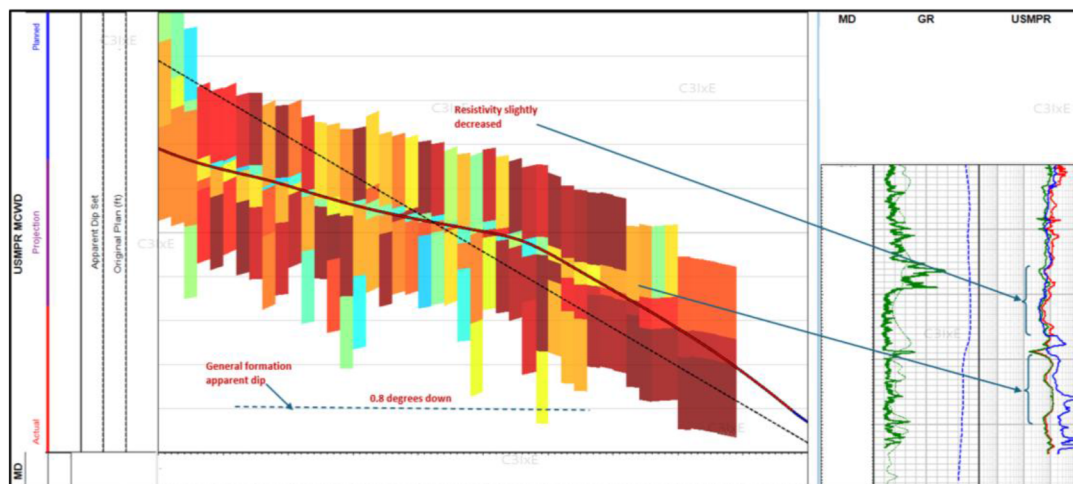


Figure 8—Real-time non-azimuthal inversion results for slant section, illustrating mapping of different resistivity layers

Drilling concluded almost as per plan, and the well successfully met the objective for this slant well within the target. The application of advanced real-time monitoring technologies with non-azimuthal inversion notably enhanced operational efficiency, minimized non-productive time, and reservoir access in the most constrained drilling environments. The non-azimuthal inversion results plotted as 3D view for better visualization with actual and planned trajectories (Figure 9).

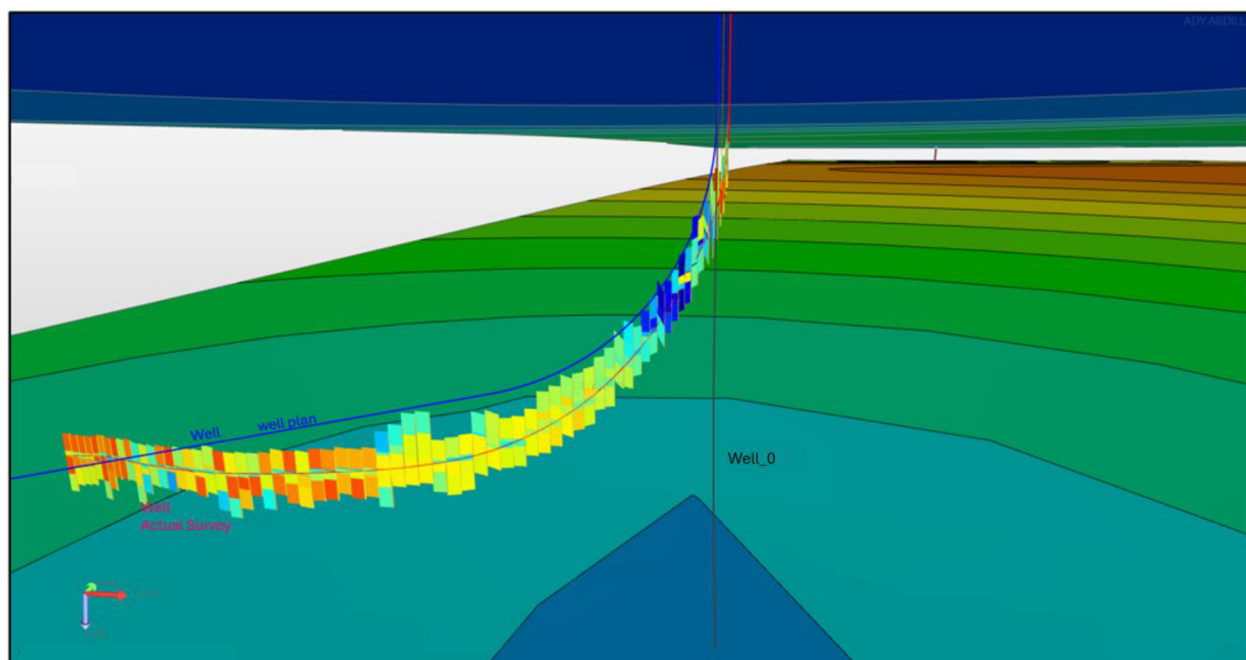


Figure 9—3D view of real-time non-azimuthal inversion results.

Conclusions

Ultra-slim multiple propagation resistivity technology, when combined with non-azimuthal inversion, provides a powerful method for resistivity-based formation evaluation in Underbalanced Coiled Tubing Drilling (UBCTD) operations. By overcoming the limitations of conventional azimuthal resistivity tools, this technology enables precise well placement, enhances reservoir characterization, and optimizes hydrocarbon recovery in challenging drilling environments.

In the well, ultra-slim multiple propagation resistivity tool with non-azimuthal inversion showed its capabilities to identify different zones based on their resistivity and detect the contrast boundaries between different zones within the tool DOI 5ft TVD in the present case based on the resistivity contrast. Additionally, it allows mapping of zones thicknesses and identification of the boundary dip along the well path and detection of the dip change as 1.2° up-dip to 0.8° down-dip in the present case.

As the industry moves toward more efficient and cost-effective drilling solutions, the adoption of ultra-slim multiple propagation resistivity and advanced inversion techniques plays a critical role in improving UBCTD success rates, reducing operational risks, and maximizing reservoir potential. UBCTD has proven to be a viable and effective method for developing the deep tight gas sandstone reservoirs. This approach with non-azimuthal inversion unlocks new potential for reservoir re-entry, tight gas redevelopment, and depleted field optimization - delivering both technical and economic value.

Calibration of the productive zone related to GR and resistivity correlation based on first deepest lateral is important when planning for multiple laterals. The first lateral results define the resistivity signature and formation dip, and the subsequent laterals would benefit from the formation dip information for prolonged interval within high productive-resistive zone. The successful deployment of these technologies in this Well provides valuable insights and a robust framework for future applications in similarly complex geological environments.

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